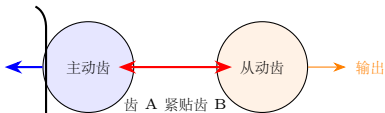
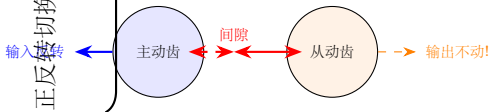


backlash = 正反转切换时主动齿空转的角度
从动齿不跟随的角度

正转（咬合）



反转（空行程/背隙）



$$E_{\text{末端}} \approx L \cdot \tan(\theta_{\text{backlash}})$$

$$L = 500\text{mm}, \theta = 0.15^\circ \Rightarrow E \approx 1.3\text{mm}$$